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Summary

Section 1 Systems of Linear Equations

1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		Comments
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2}(ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1}=a(-3)^n+bn^2+cx_n, \text{ for } n, a, b, c\in\mathbb{Z} \text{ and } n\geqslant 1.$$

Given that $x_1=1$, $x_2=-8$, $x_3=70$, and $x_4=-377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		Comments
[2]	$n=1$ $x_2=a(-3)^1+b(1)^2+cx_1$ $-8=-3a+b+c \quad ---(1)$ $n=2$ $x_3=a(-3)^2+b(2)^2+cx_2$ $70=9a+4b-8c \quad ---(2)$ $n=3$ $x_4=a(-3)^3+b(3)^2+cx_3$ $-377=-27a+9b+70c \quad ---(3)$ Using GC, $a=2,\ b=3,\ c=-5$ $x_{n+1}=2(-3)^n+3n^2-5x_n$ Using GC, $x_{10}=-7\,210\,868$ $x_{11}=36\,172\,738$	

4 *[Question: two equations, three variables – not yet sourced from the papers on the Mac]*

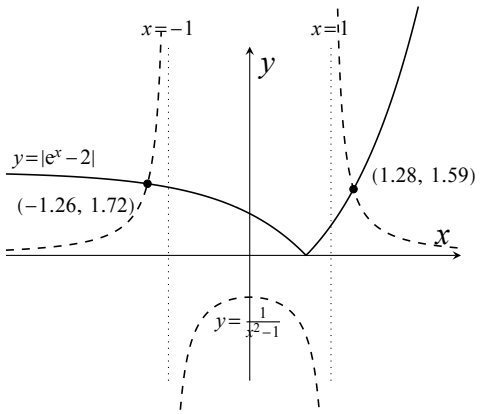
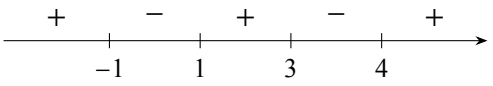
Solution		Comments
	[Solution: to follow]	

Section 2 Inequalities

5 **(a)** Solve the inequality $|e^x-2|>\dfrac{1}{x^2-1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1}\geqslant\dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

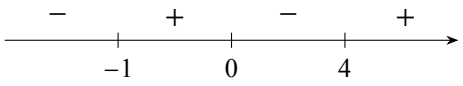
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1}-\dfrac{2}{x-3}\geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)}\geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)}\geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)}\geqslant 0$  $\therefore x\leqslant -1$ or $1<x<3$ or $x\geqslant 4$	

6 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5}>x\geqslant 3x-6.$$

[4]

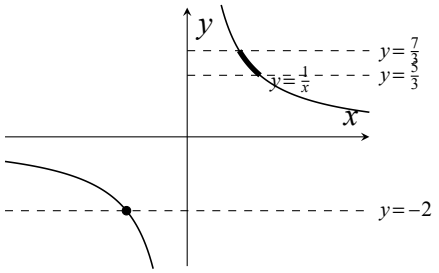
[ACJC 2026 Prelim P1 Q1]

Solution		Comments
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\dfrac{x(5x+9)}{x^2+2x+5}>x$ $5x^2+9x>x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1)\cap(2)$: $\therefore x<-1$ or $0<x\leqslant 3$	

7 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

[SAJC 2025 Prelim P2 Q1]

Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2$: $0 \leqslant 0$ (true) $x \neq -2$: $\|x+2\|>0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\frac{3}{x^2}\|1-4x^2\| = 3\left\|\frac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\frac{1+2x}{x}\right\| = \left\|\frac{1}{x}+2\right\|$ $3\left\|\left(\frac{1}{x}\right)^2-4\right\| \leqslant \left\|\frac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\frac{1}{x}=-2$ or $\frac{5}{3} \leqslant \frac{1}{x} \leqslant \frac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	